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REVIEW

Exploring advanced sentiment analysis techniques using deep learning architectures: a comprehensive survey

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Abstract
Sentiment analysis, an essential task in natural language processing, has evolved rapidly with the emergence of deep learning architectures. Although numerous studies have applied models such as CNNs, RNNs, and Transformers to sentiment classification, a comprehensive comparative synthesis across these architectures remains limited. Existing reviews often focus on specific model families or narrow application domains, leaving a gap in understanding how different deep learning paradigms collectively influence sentiment analysis performance. This paper addresses this gap by presenting a systematic review and comparative analysis of deep learning-based sentiment analysis techniques published between 2020 and 2025. The study examines major architectures—CNN, RNN, attention-based, transformer, hybrid, and graph-based models—covering their design, datasets, performance metrics, and limitations. Furthermore, it proposes a unified taxonomy of sentiment analysis models and identifies open research challenges related to data imbalance, cross-domain adaptation, multilingual processing, and model interpretability. The review aims to provide researchers with an integrated perspective on current advancements while outlining promising directions for future sentiment analysis research.

Keywords Sentiment analysis · Deep learning · Textual analysis · Neural networks · Natural language processing

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1 Introduction

With the exponential growth of user-generated content on digital platforms, extracting meaningful insights from textual data has become a priority for businesses and researchers. Sentiment analysis, also known as opinion mining, plays a pivotal role in understanding public perception and decision-making. While traditional machine learning models such as Naïve Bayes, Support Vector Machines, and Decision Trees have been widely used for sentiment classification, they often struggle with feature engineering and contextual understanding. The advent of deep learning, particularly neural network-based models, has significantly improved sentiment analysis by leveraging word embeddings and contextual representations. This paper explores various deep learning architectures applied to sentiment analysis, emphasizing their effectiveness compared to conventional techniques.

Textual sentiment analysis is an analytical method that identifies the sentiment conveyed in textual data [1]. Over the past ten years, individuals have transitioned from casual internet consumers to creators of online content. The internet will become overflowing with data as a result of this [2]. These days, judgments are based on information that is easily accessible around the clock. The digital environment is overflowing with text in this era of information explosion, originating from a multitude of sources, including social media, online reviews, news stories, and customer feedback. Human attitudes, opinions, and emotions are among the invaluable information that may be found in this vast sea of textual data [3]. Analyzing people's views, attitudes, and assessments is the primary function of textual sentiment analysis, one of the fundamental subfields of natural language processing (NLP) [4]. For example, a brand corporation may utilize textual sentiment analysis to observe how the public perceives the brand on social media, and it can promptly take appropriate action if the public's opinion of the brand suddenly becomes more unfavorable [5]. Textual sentiment analysis techniques may be broadly categorized into two groups: deep learning-based techniques and classical techniques, depending on whether the model uses deep learning technology. However, sentiment analysis faces several challenges in reaching an optimal state [6].

Sentiment expression has intrinsic complexities that vary throughout cultures, languages, and settings. This can make it difficult to appropriately read feelings that depend on context or to recognize sarcasm and irony [7]. This thorough examination attempts to shed light on the terrain of textual sentiment analysis. It reviews the development of sentiment analysis techniques over time, talks about ongoing issues, looks at novel ideas and looks at a wide range of applications in many fields. It also explores the significant influence that sentiment analysis has on decision-making procedures. This study essentially acts as a navigational aid for scholars and practitioners alike as they traverse the constantly changing field of sentiment analysis.

Despite extensive progress in deep learning-based sentiment analysis, existing review studies exhibit several limitations. Most prior surveys either concentrate on conventional neural models (e.g., CNN or LSTM) or focus narrowly on transformer-based architectures without offering a unified comparative framework. Additionally, few studies systematically evaluate model performance across diverse datasets or discuss emerging paradigms such as graph neural networks and large language models (LLMs). This fragmentation results in a lack of comprehensive understanding regarding model effectiveness, domain generalizability, and interpretability. To address these gaps, the present study offers a structured and comparative review of recent deep learning architectures for sentiment analysis, integrating empirical findings, benchmarking results, and critical discussions on methodological advancements.

While several existing reviews have examined specific categories of deep learning models for sentiment analysis, this paper advances the field by offering a unified comparative framework that synthesizes recent progress across all major deep learning paradigms–CNNs, RNNs, Transformers, hybrid, and graph-based architectures. It introduces an integrated taxonomy highlighting their methodological evolution, comparative performance, and application diversity between 2020 and 2025. Furthermore, this work identifies persistent gaps related to model interpretability, cross-domain robustness, and ethical deployment, thereby providing a consolidated foundation for guiding future research in deep learning-driven sentiment analysis.

To avoid ambiguity regarding the nature of this work, it is important to clarify that this manuscript is conceived as a comprehensive survey article rather than a comparative experiment or a position paper. Its primary objective is to systematically review, synthesize, and critically evaluate recent advancements in deep learning-based sentiment analysis, with emphasis on transformer architectures, domain-specific adaptations, multilingual modeling, and emerging learning paradigms such as zero-shot and few-shot transfer. The focus is therefore on integrating evidence from existing literature, identifying methodological trends, highlighting unresolved challenges, and outlining research gaps and future directions, rather than proposing new algorithms or presenting original experimental results.

The remainder of this paper is structured as follows. Section 2 presents a detailed review of advancements in sentiment analysis techniques, emphasizing deep learning architectures such as CNNs, RNNs, Transformers, hybrid, and graph-based models, with focused discussions on specialized models like DialogueRNN, BiERU, GCNNs, and Deep Belief Networks. Section 3 presents the mathematical foundations of major deep learning architectures used in sentiment analysis. Section 4 outlines the methodology adopted for conducting this systematic review, including the search strategy, selection criteria, data extraction, and analytical framework used for comparative evaluation. Section 5 provides performance analysis and synthesizes findings from the reviewed studies, highlighting comparative strengths and weaknesses of different deep learning models. Section 6 discusses the major challenges in sentiment analysis, including data scarcity, interpretability, and cross-domain adaptation. Section 7 identifies key research gaps and underexplored areas requiring further investigation. Section 8 presents future research directions, emphasizing model interpretability, multimodal integration, and ethical considerations. Section 9 concludes the paper by summarizing key insights and reaffirming the role of deep learning in advancing sentiment analysis. Finally, Section 10 provides declarations regarding authorship, funding, and conflict of interest.

2 Advancements in sentiment analysis technique

Early sentiment analysis approaches relied on lexicon-based methods and machine learning models, which required extensive feature engineering. Recent advancements in deep learning have led to the adoption of architectures such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), long short-term memory (LSTM) networks, and transformer-based models like BERT and GPT, as given in Fig 1.

Recent studies consistently demonstrate that deep learning architectures significantly outperform traditional machine learning methods such as Naïve Bayes, SVM, and logistic regression in sentiment analysis tasks across multiple domains. For example, Khan et al. [8] showed that transformer-based models such as BERT and RoBERTa achieved up to 15% higher F1-scores compared to conventional classifiers on large-scale social media datasets.

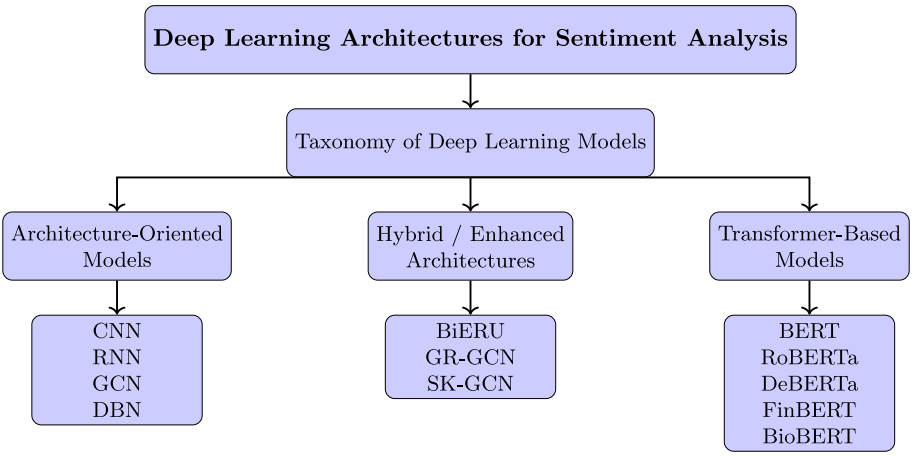


Fig. 1 Conceptual taxonomy framework categorizing deep learning architectures for sentiment analysis

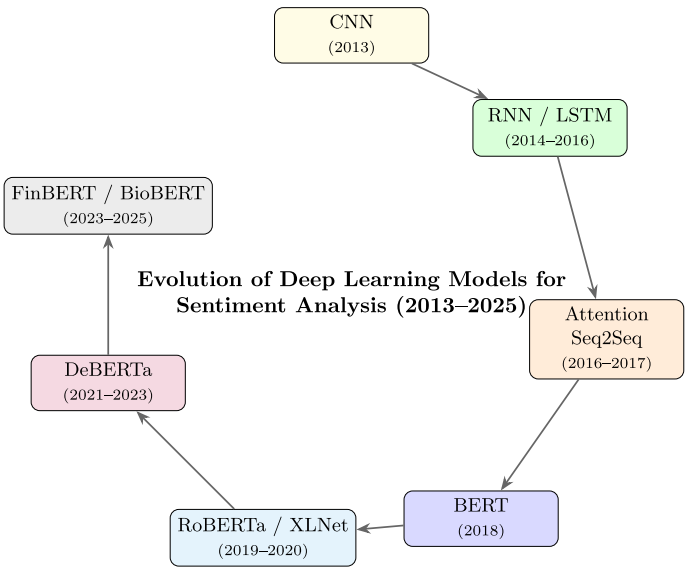


Fig. 2 Evolution of sentiment analysis models from CNN and RNN to modern transformer-based architectures

Similarly, Cao [9] and Kaur et al. [10] reported that DeBERTa and RoBERTa outperform traditional baselines and earlier neural models in cross-domain sentiment classification, confirming the robustness of deep contextual representations over hand-crafted feature approaches. This section critically examines existing research and highlights key developments in sentiment analysis techniques. Evolution of sentiment analysis models from CNN and RNN to modern transformer-based architectures is given in Fig. 2. The diagram emphasizes chronological progression and model interconnectivity.

Previous research in sentiment analysis can be segmented into three phases: lexicon-based approaches, statistical machine learning models, and deep learning. While lexicon methods lack adaptability and learning capability, statistical classifiers struggle with semantic

ambiguity. Deep learning models such as CNNs offer spatial feature learning, RNNs handle temporal dependencies, and Transformers revolutionize contextual understanding via self-attention mechanisms.

With an emphasis on deep learning models, the literature review offers a perceptive examination of current developments in sentiment analysis methods. It starts by introducing DialogueRNN, a model built on a hierarchical recurrent neural network (RNN) architecture that is intended to comprehend textual utterances in conversational scenarios. The performance of BiERU, a sentiment analysis system that combines ensemble learning techniques with bidirectional long short-term memory (BiLSTM) networks, is then evaluated in the review. The use of graph convolutional neural networks (GCNNs) in language processing tasks is next covered, with a focus on how well these networks can capture complex linguistic linkages and hierarchical structures. In addition, the article looks at how recurrent neural networks (RNNs) are used in different fields for sentiment analysis. Finally, it examines how well Deep Belief Networks (DBNs) perform sentiment analysis tasks, emphasizing how well they can identify intricate patterns in textual material.

2.1 The depiction of textual expressions in DialogueRNN

DialogueRNN is used to describe a model that is designed to comprehend textual expressions in talks. The hierarchical recurrent neural network (RNN) architecture is used in this model to address the complex dynamics of emotion, agreement, and disagreement. The architectural design incorporates utterance-level recurrent neural networks (RNNs) to encode individual remarks, while a context-level RNN is responsible for capturing the larger conversational context. By conducting thorough experiments on different datasets of conversations, DialogueRNN demonstrates exceptional performance in tasks such as identifying emotions and classifying dialogue acts. This confirms its ability to accurately interpret textual expressions within conversational situations. Some of the recent research works related to the DialogueRNN were reviewed in this section. Simplified architecture of DialogueRNN given in Fig. 3 showing three interconnected GRU components: the Global GRU maintains overall dialogue context, the Party GRU tracks speaker-specific states, and the Emotion GRU models emotional evolution across utterances.

In 2020, Wen and Young [11] introduced RNNLG, a statistical natural language generator based on recurrent neural networks (RNNs), which can produce utterances from dialogue act pairings without specified syntax or semantic alignments. A rigorous comparison of the main RNN-based NLG models is presented. The second contribution adapts models from one domain to another to evaluate the system's scalability. By pairing RNN-based NLG models with a proposed data counterfeiting approach and a discriminative objective function, a pre-trained model may be swiftly adapted to different domains with few examples. The first RNN-driven dialogue system testing tool was designed and implemented DialTest by Liu et al. [12]. DialTest uses transformation operators to realistically update seed data while keeping Oracle information. DialTest uses Gini impurity to drive test generation to improve fault detection. Validate DialTest with several experiments. First, test it on two core natural language comprehension tasks: intent detection and slot filling. According to the experiment, DialTest can detect hundreds of incorrect behaviors for RNN-driven natural language understanding (NLU) modules in dialogue systems and enhance their performance by retraining with the generated data. Also, do a case study on an industrial dialogue system to evaluate DialTest in real-world use. DialTest detects faults and strengthens RNN-driven dialogue systems, according to the study.

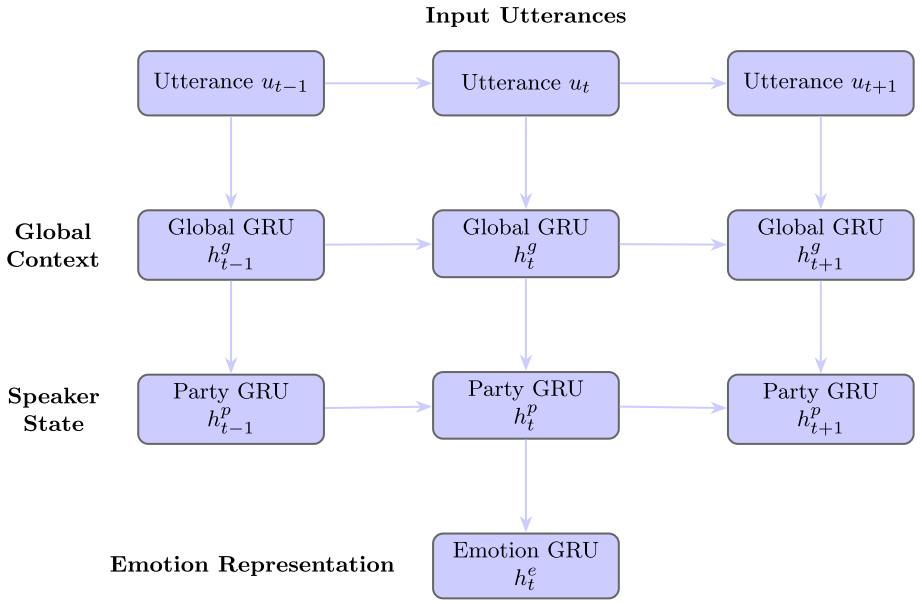


Fig. 3 Simplified architecture of DialogueRNN

Multi-turn dialogues from a large corpus using context and subject information to generate more coherent context-sensitive responses were built by Sun and Ding [13] in 2022. We augment previous models and attention mechanisms and propose the HAT model, a hierarchical model to better solve dialogue context. This strategy gives the model more contextual information while processing and increases its contextual relevance and response quality. We also pre-train the LDA (Latent Dirichlet Allocation) topic model to extract topic words from dialogue material and maintain as much subject information as feasible to address the lack of topics in the responses. Experiments in numerous corpora show that our model outperforms most hierarchical and non-hierarchical models on multiple assessment metrics. In 2023, Yang et al. [14] explored a model from collecting the speaker's intention and crucial discourse information for complicated discourse queries, resulting in incorrect replies. Utilize the relational graph convolutional network (R-GCN) to structure dialogue information and develop a speaker prediction challenge to improve discourse logic capture. Also, build a key utterance information decoupling module that filters noise and focuses on question-related discourse information flow. Many FriendsQA and Molweni studies show that the approach beats competing baselines and current state-of-the-art models, especially with more rounds of dialogue and questions about individuals, events, and time. Ghosal et al. [15] examined and quantified the role of context for emotion, intent, and dialogue act identification utilizing state-of-the-art dialog comprehension approaches in 2020. In particular, modify utterance contexts and explore their effects on tasks and baselines. This shows the contextual factors that govern distinct dialogue features. This can improve dialogue understanding models and boost future text creation methods. Table 1 demonstrates the research gaps of aim, method, advantage, disadvantage, and Findings from the existing works of DialogueRNN.

Table 1 Research gaps from the existing works of DialogueRNN

Author name	Aim	Methods	Advantage	Disadvantage	Findings
Wen and Young [11]	Presenting RNNLG for natural language generation, and proving scalability through domain adaption	Statistic natural language production using RNN	Learning to construct utterances from dialogue act utterance pairs without syntaxes or semantic alignments	May demand lots of processing power to train RNN models	With corpus-based and human evaluations
Liu et al. [12]	Creating DialTest for RNN-driven dialogue systems	Gini impurity for test generation, and transformation operators	Effectively detects RNN-driven dialogue system errors	Specific dialogue system architectures may require fine-tuning	Extensively tested on intent detection, slot filling, and industrial dialogue systems
Sun and Ding [13]	Proposing context-sensitive multi-turn dialogue generation using HAT	Topic extraction with hierarchical attention and pre-training LDA	Uses discourse context and topic to respond coherently	Hierarchical structure may increase complexity	Outperformed prior models in contextual relevance and response quality
Yang et al. [14]	Multi-party dialogue comprehension improved by dialogue logic graph module and speaker prediction job	Relations graph convolutional network (R-GCN) for dialogue logic graph, speaker prediction, and important utterance information decoupling	Creates dialogue logic graph module for dialogue organization	Dialogue logic graph module construction and processing may take more computational resources	Handles multi-round discussions and inquiries about persons, events, and time better than competitor baselines and state-of-the-art models
Ghosal et al. [15]	Finding out how context affects emotion, intent, and dialogue act identification	Distorting discussion context with perturbations, using state-of-the-art dialog understanding algorithms as baselines	Explains contextual controls for dialogue comprehension	The context-distorting perturbations used may affect results	Measures context's impact on emotion, intent, and discourse act identification

2.2 Assessment of BiERU's performance in sentiment analysis

It examines how well the sentiment analysis system Bidirectional Emotional Recurrent Unit (BiERU) performs. Bidirectional long short-term memory (BiLSTM) networks and ensemble learning methods are combined in BiERU. BiERU's performance is fully evaluated by extensive testing on benchmark sentiment analysis datasets, which include sentiment polarity classification and fine-grained sentiment analysis tasks. Metrics including accuracy, precision, recall, and F1-score are examined in the evaluation to determine how effective the system is. Results reveal that BiERU performs competitively in sentiment analysis tasks, demonstrating its promise as a reliable method for identifying sentiment in textual data in a range of contexts and domains.

A fast, compact, and parameter-efficient party-ignorant framework named the bidirectional emotional recurrent unit for conversational sentiment analysis was introduced by Li et al. [16] in 2022. In this system, a generalized neural tensor block followed by a two-channel classifier is designed to perform context compositionality and sentiment classification, respectively. Extensive experiments on three standard datasets demonstrate that our model outperforms the state of the art in most cases.

In 2023, Bordoloi and Biswas [17] covered strategies, algorithms, and other variables for constructing an effective sentiment analysis model in detail. The study critically evaluates sentiment analysis framework modules and discusses method or system flaws. Based on data, the report presents transdisciplinary sentiment analysis applications and research paths.

A new Matching feature with an output label-based advanced technique (MAT) to initialize ANN weights may solve the problem was suggested by Thakkar et al. [18] in 2022. Simulation findings show that PPR+MAT-ANN outperforms EPR+R-ANN, which integrates preprocessing (EPR) stages with R-ANN. PPR+MAT-ANN architecture is simpler than the NeuroSent tool and performs better on the Dranziera protocol.

Aspect-based sentiment Analysis (ABSA) permits users to compute the sentiment for an aspect in a specific context was suggested in 2022 by Datta and Chakrabarti [19]. The domain-specific knowledge plays a major role in analyzing the sentiments of aspects. This study examined whether knowledge-driven ABSAs can improve efficiency and effectiveness. These issues can be addressed by comparing machine learning algorithms using movie data. It analyzes sentiment analysis on movie datasets to compare classifiers like SVM, DT, NB, KNN, and NN. Extraction of significant features and aspects from contexts can increase classification accuracy. Movie data preprocessing includes stop words, punctuation, lowercase conversion, and stemming. Opinion words are collected in as "pect" extraction phase "e", and features are retrieved using polarity score and word2vector. The collected features are fed to machine learning techniques like K-nearest neighbor (KNN), decision tree (DT), Naive Bayes (NB), support vector machine (SVM), and neural network. Experimental results and in-depth analysis show that the five techniques focus more on performance improvement.

In 2021, Tam et al. [20] proposed a CNN-Bi-LSTM model integration. ConvBiLSTM translates tweets into numerical values, the CNN layer gets feature embedding as input and produces smaller features, and Bi-LSTM classifies. Word2Vec and GloVe were used to see how word embedding affected the model. Tweets and SST-2 datasets were processed with ConvBiLSTM. Using 91.13% accuracy, the ConvBiLSTM model using Word2Vec on retrieved Tweets dataset surpassed the others.

A hybrid uses a convolutional neural network (ConVnet) architecture to merge two LSTM blocks (BiLSTM) with the other direction of propagation to classify brain signal data from electroencephalogram (EEG) on individuals, typically development (TD) and autism (ASD) from the Simon Foundation Autism Research Initiative (SFARI) database was proposed by

Ali et al. [21] in 2022. With a 70:30 data distribution, accuracy was 97.7%. The results showed that proposed autism classification approaches outperformed the current state-of-the-art and could contribute to neuroscience research.

Lin et al. [22] introduced StyleBERT, a text-audio sentiment analysis framework that enhances emotional information in unimodal representations by learning modality styles in 2023. This results in an effective representation before fusion, reducing the need for fusion. Proposed a Bidirectional Style Enhancement module that learns contextualized and differentiated style representations for each modality, capturing semantic information and modality-specific characteristics. Use solely Log-Mel spectrograms as audio modality inputs and encode them using a multi-head self-attention technique to learn fine-grained acoustic representation. The proposed StyleBERT multimodal framework beats state-of-the-art multimodal baselines in comprehensive experimental results on three frequently used benchmark datasets. Table 2 demonstrates the research gaps of aim, method, advantage, disadvantage, and Findings from the existing works of BiERU. A novel dataset for sentiment analysis experiments is also provided by Manoj Singh [23], in which more then 10,000 records are available [24].

Architectural overview

BiERU consists of two parallel recurrent networks that process textual input in opposite directions. The hidden states from both directions are fused through an emotion-aware gating mechanism, allowing the model to dynamically regulate emotional context. The forward and backward contextual representations are concatenated and modulated by an emotion vector, as described by:

$$h_t = \tanh\left(W_f[\overrightarrow{h_t}; \overleftarrow{h_t}] + W_e e_t + b\right), \tag{1}$$

where $\overrightarrow{h_t}$ and $\overleftarrow{h_t}$ denote the hidden states from the forward and backward recurrent units, respectively, e_t is the emotion embedding at time step t, and W_f, W_e, b are trainable parameters. The resulting hidden state h_t thus encodes both bidirectional semantic context and emotion-specific cues. Figure 4 shows the forward and backward recurrent flows integrated through an emotion-aware gating mechanism to produce a bidirectional sentiment representation.

Analytical significance

The key innovation of BiERU lies in its explicit modeling of emotional flow through the sequential structure of text. By integrating an emotion-aware gating mechanism, BiERU not only captures word-level sentiment but also accounts for emotional dependencies across sentences or utterances. This capability has been shown to improve contextual sentiment tracking in multi-turn dialogues where emotional polarity may shift between speakers. Comparative studies have demonstrated that BiERU consistently outperforms baseline RNNs and LSTMs on emotion-rich datasets such as IEMOCAP and DailyDialog, validating the contribution of bidirectional emotional context modeling [16, 25].

Table 2 Research gaps from the existing works of BiERU

Author name	Aim	Methods	Advantage	Disadvantage	Findings
Li et al. [16]	A concise and efficient contextual conversational sentiment analysis framework	BiERU, neural tensor block, two-channel classifier	For context compositionality and sentiment classification, use a neural tensor block and a two-channel classifier	May miss speaker-specific sentiments	Performs better than top models on three datasets
Bordoloi and Biswas [17]	Systematically explaining sentiment analysis model development	Review and critique of sentiment analysis frameworks and methodologies	Highlights flaws and suggests research directions	May not provide inexperienced practitioners with implementation information	Discusses sentiment analysis methods and studies
Thakkar et al. [18]	A unique neural network weight initialization method improves sentiment analysis	MAT and ANN	Improved performance with innovative weight initialization	Must carefully tune preprocessing and weight initialization values	Performs better than current methods
Datta and Chakrabarti [19]	Using machine learning methods for movie aspect-based sentiment analysis	Data prep, feature extraction, and ML algorithms (SVM, DT, NB, KNN, NN)	Compares sentiment classification machine learning algorithms	Quality of feature extraction and dataset preprocessing affects performance	Stresses feature extraction and machine learning methods for sentiment analysis
Tam et al. [20]	An integrated CNN-Bi-LSTM model for Twitter sentiment analysis	CNN, Bi-LSTM, Word2Vec, GloVe	Proves the model works on Twitter sentiment analysis	May need model hyperparameter tweaking for best performance	Better than CNN and Bi-LSTM Twitter sentiment analysis algorithms
Ali et al. [21]	An EEG-based hybrid deep learning model for autism Categorization	ConVnet, BiLSTM, EEG data analysis	Classifies Autism Spectrum Disorder (ASD) using EEG data automatically	Data collection and model implementation may require skill	Better ASD classification than current methods
Lin et al. [22]	The StyleBERT model for multimodal sentiment analysis with effective modality representation learning	Bidirectional Style Enhancement, multi-head self-attention	Improves unimodal emotional information	Multimodal model training may demand more computational resources	Performs better than multimodal benchmark baselines

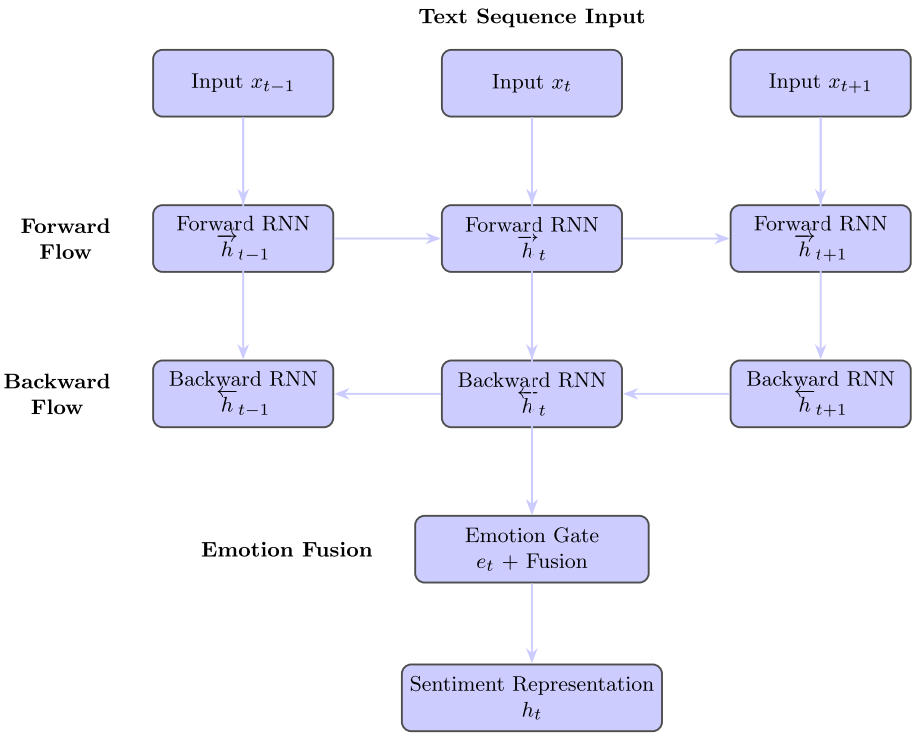


Fig. 4 Simplified architecture of the BiERU model

Limitations and research scope

Despite its superior performance, BiERU also presents certain limitations. The dual recurrent structure increases computational cost and memory usage, making it less efficient for large-scale datasets. Moreover, the model's performance is sensitive to the quality of emotion embeddings; inaccurate or biased representations can lead to degraded sentiment detection accuracy. Future improvements could integrate transformer-based attention modules or domain-adaptive emotion encoders to enhance contextual alignment and scalability.

Overall, BiERU bridges the gap between semantic and affective understanding by explicitly incorporating emotion-aware recurrence into bidirectional text modeling. Its balanced integration of semantic and emotional signals makes it an important intermediate step between conventional LSTM models and modern attention-based sentiment architectures.

2.3 Language processing with graph convolutional neural networks

Graph Convolutional Neural Networks (GCNNs) utilize graph topologies to depict linguistic data, with nodes signifying words or tokens and edges representing syntactic or semantic connections. This research examines the superior ability of GCNNs to capture intricate language connections and hierarchical structures in comparison to conventional NLP models. GCNNs show potential in many NLP tasks, including text classification, semantic role labeling, and sentiment analysis, by combining graph-based representations with

neural processes. This study emphasizes the potential of Graph Convolutional Neural Networks (GCNNs) in enhancing language processing tasks through the utilization of abundant structure information.

The fundamentals of GCNs and their applications in these two sectors are thoroughly reviewed by Ren et al. [26] in 2022. First, provide an overview of the fundamental ideas behind the two categories of GCNs: spectral and spatial approaches. Next, categorize GCN applications into two groups: computer vision and natural language processing. Then go into detail about many apps from each category. Lastly, describe the shortcomings of GCNs and speculate about potential future research paths.

Tayal et al. [27] suggested a Graph-regularized Graph Convolution Network (GR-GCN) in 2020, which enhances graph convolution networks by adding label dependencies in the output space, to address the shortcomings of textual characteristics in a short text. On both private and external datasets, the model produces state-of-the-art results, surpassing numerous baseline approaches by as much as 6%. Additionally, demonstrates that GR-GCN is more resilient to noise in textual characteristics than baseline techniques.

In 2021, Du et al. [28] used the widely used Graph Convolutional Network (GCN) and word dependency trees to integrate syntactical restrictions and long-range word dependencies into the sentence encoding process. The adjacency matrix is created by existing syntax-aware GCN algorithms based on the dependency tree's connection between two words. However, the word dependency type which represents the way words are connected in dependency trees is not modeled by them. They are unable to discern between the various contributions made by various word-dependence routes. Propose a GCN version extended by a new Word Dependency Gate method, to prevent adding unnecessary word dependencies that degrade language understanding. Depending on the type of word dependency and its context, the Word Dependency Gate can adaptively maintain the balance between the inclusion and exclusion of particular word dependency paths. Test results demonstrate that our method can successfully integrate the pertinent syntactical dependence in BERT and attain a cutting-edge result in the End-to-End Aspect-Based Sentiment Analysis and Relation Triple Extraction assignments.

Zhao et al. [29] introduced three architectures of the Multi-head Pooling-based Graph Convolutional Network (MP-GCN) for semi-supervised short text classification, which focuses on the node representation learning of 1-order, 1 and 2-order, and 1-order heterogeneous graphs, respectively, to address these issues. Pre-training word embedding as the initial node feature is not required; instead, it solely concentrates on the text graph's structural information. The multi-head technique is utilized to give different representation subspaces for pooling without adding trainable parameters, and a graph pooling based on self-attention is presented to evaluate and pick significant nodes. Based on experimental results, MP-GCN beats state-of-the-art models across five benchmark datasets without the use of pre-training embedding.

A novel joint model that integrates the syntactic structure for learning slot filling and intent detection concurrently, using a graph convolutional network over dependency trees was suggested by He et al. [30] in 2020. Based on experimental results, our suggested model outperforms previous work and reaches state-of-the-art performance on two public benchmark datasets. Finally, we use the BERT model to enhance the performance even further for intent recognition and slot filling. Pereira and de [31] suggested a novel approach to create a text graph that can impact the TGCN's outcome and result in lower memory usage by examining the outcomes of text representations in 2022. Thus, the application of graph neural networks (GNN) for text processing has been investigated by several academics. Because global and contextual information can be captured in a corpus, GNNs do exceptionally well in text categorization tasks. Benchmark classification trials demonstrated the Text Graph

Convolutional Network's (TGCN) ability to outperform conventional NLP approaches. But when it comes to building text graphs, this approach uses a lot of memory

In 2020, a new Syntax- and information-based Graph Convolutional Network (SK-GCN) model for aspect-level sentiment categorization that makes use of GCN to utilize commonsense information and the syntactic dependency tree was proposed by Zhou et al. [29]. Specifically, develop two methodologies, SK-GCN1 and SK-GCN2, to model the syntactic dependency tree and commonsense knowledge graph, respectively, to improve the representation of the phrase toward the given aspect. Syntax-based GCN (S-GCN) and Knowledge-based GCN (K-GCN) are used by SK-GCN1 to model the dependency tree and knowledge graph independently, while SK-GCN2 models them together. Additionally, we use pre-trained BERT for this task, achieving new state-of-the-art outcomes. In comparison to state-of-the-art techniques, extensive trials on five benchmark datasets show that the methodology can effectively increase the performance of aspect-level sentiment classification. Table 3 demonstrates the research gaps of aim, method, advantage, disadvantage, and Findings from the existing works of GCN.

Architectural overview

A typical GCNN model represents a sentence as a graph $G = (V, E)$, where each node $v_i \in V$ corresponds to a word embedding and each edge $(v_i, v_j) \in E$ encodes a syntactic or semantic dependency. The graph convolution operation updates node representations by aggregating information from their neighbors according to:

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right), \tag{2}$$

where $\tilde{A} = A + I$ is the adjacency matrix with self-connections, $\tilde{D}$ is the degree matrix, $H^{(l)}$ denotes node embeddings at layer l, $W^{(l)}$ are trainable weights, and $\sigma(\cdot)$ is a nonlinear activation (e.g., ReLU). This operation allows each node (word) to iteratively refine its representation by aggregating contextual information from syntactically related words. Figure 5 shows Simplified architecture of a Graph Convolutional Neural Network (GCNN) for sentiment analysis. Words are represented as nodes connected by syntactic dependencies, and multiple graph convolution layers aggregate and refine contextual features before classification.

Analytical insights

The advantage of GCNNs in sentiment analysis lies in their ability to model long-range dependencies and hierarchical sentence structures that are often overlooked by sequential models. For example, in the sentence "The food was great but the service was terrible," the sentiment polarity of "food" and "service" are connected via a syntactic contrast ("but"), which a GCNN can effectively capture through dependency-based edge connections. This structural representation enables GCNNs to outperform CNNs and RNNs on aspect-based sentiment classification (ABSA) and fine-grained opinion mining tasks [32, 33].

Limitations and future research

Although GCNNs provide strong structural modeling capabilities, they exhibit certain drawbacks:

Table 3 Research gaps from the existing works of GCN

Author name	Aim	Methods	Advantage	Disadvantage	Findings
Ren et al. [26]	GCNs in natural language processing and computer vision: a comprehensive overview	GCN theory and application literature review	Explains research patterns and future directions	Possibly lacking GCN architecture study	GCN principles and applications in NLP and computer vision
Tayal et al. [27]	Introduction of GR-GCN for brief text classification	The graph-regularized graph convolution network	Gets state-of-the-art textual feature results and noise resistance	May need hyperparameter adjustment for best performance	Up to 6% faster than baseline approaches on brief text classification
Du et al. [28]	Syntactical constraints improve sentence encoding using Graph Convolutional Networks	Word Dependency Gate, Graph Convolutional Networks	New Word Dependency Gate mechanism for GCNs	This may complicate syntactical dependency modeling	Outstanding End-to-End Aspect-Based Sentiment Analysis and Relation Triple Extraction Performance
Zhao et al. [29]	Introducing MP-GCN for semi-supervised short text categorization	Graph Convolutional Network with Multi-head Pooling	State-of-the-art performances on five benchmark datasets without pre-training word embeddings	May need careful architectural and hyperparameter selection for best performance	Without pre-training word embeddings, outperforms state-of-the-art algorithms on numerous benchmark datasets

Table 3 continued

Author name	Aim	Methods	Advantage	Disadvantage	Findings
He et al. [34]	A GCN-based joint model for slot filling and intent detection in spoken language interpretation	Graph Convolutional Networks, BERT	Increases slot filling and intent detection performance with syntactic structure	Processing syntactic relationships may take more computing	Outstanding slot filling and intent detection performance
Pereira and de [31]	Introducing memory-efficient Text categorization with TGCN	TGCN text graph convolutional network	Reduces memory use in Text Graph Convolutional Networks (TGCNs) for text classification	Optimizing memory efficiency may require careful parameter selection	Improves text classification while saving memory
Zhou et al. [32]	Proposing SK-GCN for aspect-level sentiment classification using syntax and knowledge	Pre-trained BERT model using SK-GCN	Improves aspect-level sentiment classification with syntactic dependency tree and commonsense	Syntactic and commonsense knowledge integration may require preprocessing	Provides better aspect-level sentiment classification than current approaches

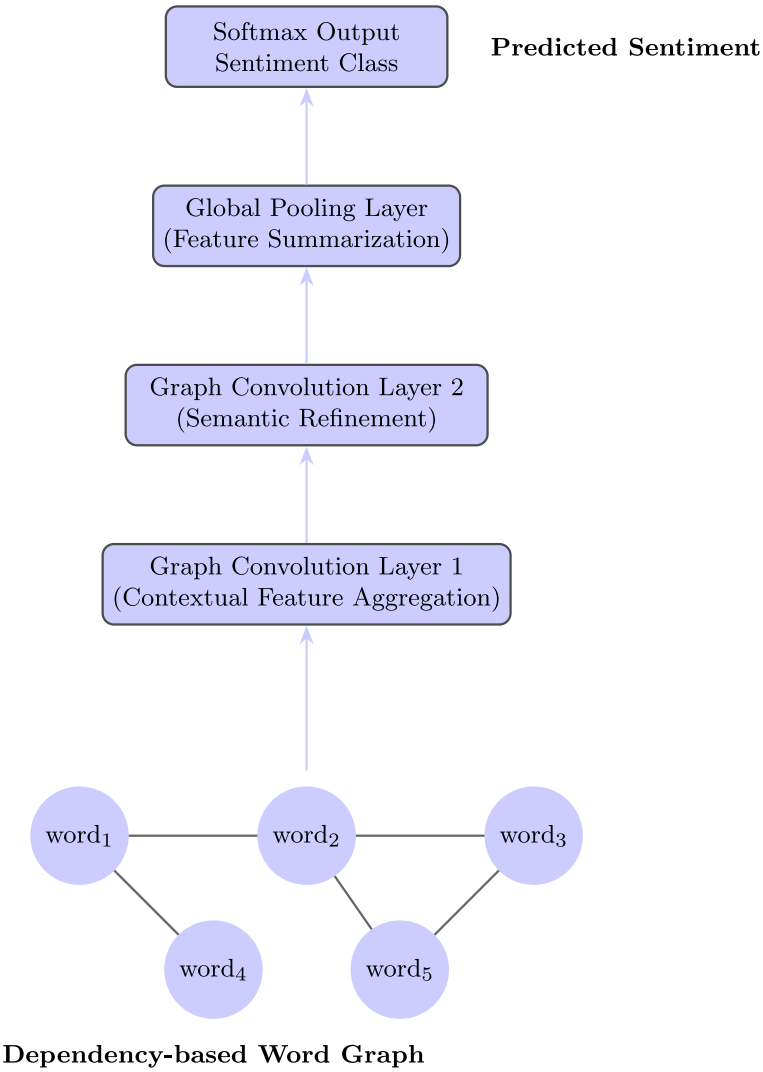


Fig. 5 Simplified architecture of GCNN for sentiment analysis

- *High computational complexity* Constructing and training graph representations for long documents is computationally expensive.
- *Graph dependency* The performance of GCNNs heavily depends on the quality of the syntactic parser used to generate graph edges.
- *Limited scalability* Sentence-level graphs are effective, but document-level graph modeling remains challenging due to memory and sparsity constraints.

Recent advancements in hybrid models (e.g., SK-GCN and GR-GCN) aim to integrate graph reasoning with attention mechanisms, achieving better context fusion and improved generalization across domains.

2.4 Applications of RNNs in sentiment analysis

Recurrent neural networks (RNNs) are highly proficient at collecting sequential information, enabling them to effectively represent dependencies within words. Sentiment analysis tasks include the utilization of these entities to analyze textual inputs, encode contextual information, and make predictions regarding sentiment labels. Recurrent neural networks (RNNs) are capable of effectively managing variable-length inputs and addressing the issue of vanishing gradients by employing techniques such as long short-term memory (LSTM) and gated recurrent unit (GRU). Their utilization in sentiment analysis encompasses many fields, such as monitoring sentiment on social media, analyzing product reviews, and mining opinions, demonstrating their adaptability and effectiveness in extracting sentiment from textual data.

An important class of deep neural networks is made up of recurrent neural networks (RNNs) and their architectural variants, such as gated recurrent units (GRU) and long short-term memory (LSTM), which are primarily designed by Tembhurne and Diwan [35] for feature extraction from temporal and sequential inputs in 2021. Critically examine the function of sequential deep neural networks in sentiment analysis of multimodal data. Input to SA and related tasks might be visual, textual, auditory, or any combination of these. Sequential deep neural networks have an intrinsic sequential structure. In particular, provides a thorough analysis of the applicability, difficulties, problems, and strategies for multimodal, textual, and visual SA utilizing RNN and its architectural variations.

In 2021, Basiri et al. [36] suggested the CNN-RNN Bidirectional Attention-based Deep Model (ABCDM). ABCDM considers temporal information flow in both directions and extracts both past and future contexts by using two independent bidirectional LSTM and GRU layers. Additionally, the outputs of the bidirectional ABCDM layers are subjected to the attention mechanism, which highlights various words to varying degrees. ABCDM uses pooling and convolution processes to extract position-invariant local features and minimize the dimensionality of the data. Sentiment polarity identification, the most common and important activity of sentiment analysis, is used to assess the efficacy of ABCDM. Three Twitter datasets and five review datasets were used for the experiments. When sentiment analysis is conducted, ABCDM outperforms six recently suggested DNNs, achieving state-of-the-art results on both lengthy review and short tweet polarity categorization.

A huge volume of data was produced by social networking sites, blogs, forums, and opinion sites. Opinions, feelings, and beliefs regarding politics, goods, and services comprise this type of unstructured data was explored by Pathak et al. [37] in 2020. Politicians, corporate sectors, and end users are all greatly impacted by the opinions shared on social media. Thus, automated sentiment analysis is required for extracting, evaluating, summarizing, and forecasting sentiments from massive amounts of unstructured data. Sentiment analysis is an automated method that separates opinions from data and categorizes them into three groups: neutral, negative, and positive. One of the major issues in natural language processing is the scarcity of labeled data for sentiment analysis. Because deep learning models naturally facilitate automated and hierarchical learning, deep learning has emerged as one of the most sought-after ways to tackle this problem. It tries to clarify the importance of deep learning approaches for sentiment analysis and presents a taxonomy of qualities to be taken into account for deep learning-based sentiment analysis, with a focus on applying these approaches to sentiment analysis.

In 2021, the enormous volume of multilingual text data must be categorized to carry out sentiment analysis (SA), which was proposed by Baliyan et al. [38]. Since it is difficult to find labeled data for multilingual text classification, a labeling technique is proposed here using a large amount of labeled English datasets via Neural Machine Translation (NMT). The

suggested model would produce context and classify text into sentiment classes by feeding Recurrent Neural Network-Long Short-Term Memory (RNN-LSTM) with Global Vectors (GloVe) as word embeddings. The primary feature is the labeling system, which addresses the challenge of gathering tagged multilingual text data. Depending on the text's, documents, or series of texts' content, the suggested model can also categorize sentiment analysis.

Several deep learning techniques to precisely forecast customer opinions based on mobile phone evaluations were developed by Alharbi et al. [39] that were sourced from Amazon.com in 2021. Based on an analysis of these reviews and a classification of them as neutral, negative, or favorable, the prediction was made. A variety of deep learning algorithms, including the basic RNN and its four variations–Long Short-Term Memory Networks (LRNN), Group Long Short-Term Memory Networks (GLRNN), Gated Recurrent Unit (GRNN), and Update Recurrent Unit (UGRNN)–have been put into practice and assessed. All of the assessed algorithms–including Glove, word2vec, and FastText by Skip-grams–are coupled with word embedding as a feature extraction method for sentiment analysis. Accuracy, recall, precision, and F1-score are the metrics used to assess the five distinct algorithms using the three feature extraction techniques on both balanced and unbalanced datasets. The GLRNN methods using FastText feature extraction were found to have the maximum accuracy of 93.75% for the unbalanced dataset. When compared to other approaches described in the literature, our result yielded the highest accuracy on this dataset. The LRNN algorithm's best accuracy for the balanced dataset was 88.39%.

Arabic-language material on websites and social media has increased dramatically in the past ten years were evaluated by Alhumoud and Wazrah [40] in 2022. Since opinions are freely and publicly expressed on social media, this makes them an excellent resource for trend analyses, which are carried out using more traditional techniques for interpreting language, such as sentiment analysis. Deep learning has been used more and more to test opinions because of its accuracy in analyzing unstructured data. Large morphological variations are exhibited by recurrent neural networks (RNNs), which show great promise in textual analysis. RNNs were employed in 193 studies for sentiment analysis in the English language and 24 studies for sentiment analysis in the Arabic language. The fields that the research covered, the strengths and limitations of the models, and the quantity and quality of the datasets for various dialects that were available differed. To discover, categorize, and assess state-of-the-art studies employing RNNs for Arabic sentiment analysis, this work conducts a systematic review of the literature. These variations are worthy of notice and observation.

An extensive description of the process for accomplishing and the uses of sentiment analysis were proposed by Wankhade et al. [4] in 2022. It then assesses, contrasts, and examines the methods employed to fully comprehend the benefits and drawbacks of each. To determine future paths, the sentiment analysis issues are finally explored. The quick development of web-based applications, such as blogs and social media sites, has led to evaluations and comments about daily life. Sentiment analysis is the process of collecting and examining people's views on a range of issues, goods, and services. Businesses, governments, and individuals may all benefit from people's opinions when gathering data and formulating their ideas as the basis for choices.

Sachin et al. [41] made an effort to survey the many deep learning methods that have been used for sentiment analysis and categorization in 2020. Using an Amazon review dataset, have applied the baseline models for LSTM, GRU, Bi-LSTM, and Bi-GRU. Sentiment analysis of text is a significant and difficult task. One issue that businesses frequently deal with is sentiment analysis of consumer evaluations.

Cen et al. [42] presented three deep learning networks used for sentiment analysis of IMDB movie reviews in 2020. 50% of the dataset's reviews were positive and 50% were negative.

While convolutional neural networks (CNN) are frequently employed in image recognition, recurrent neural networks (RNN) and long short-term memory (LSTM) neural networks are the two primary varieties that are frequently utilized in NLP tasks. The outcomes demonstrate that the CNN network model, when used for sentiment analysis of movie reviews, may produce good classification results. While RNN and LSTM recorded an accuracy of 68.64% and 85.32%, respectively, CNN reported an accuracy of 88.22%. Patel et al. [43] implemented the RNN deep learning-based classification method, evaluated the classifier's performance using the pre-processed data, and achieved an accuracy of 94.61% in 2020. In this case, the RNN method was utilized rather than the machine learning algorithm because the latter only operates on a single layer, while the former operates on several layers and produces superior results. In 2020, Li et al. [22] provided a general end-to-end segmentation model, called SEGBOT, given these constraints. It begins by encoding an input text sequence using a bidirectional recurrent neural network. Next, to choose text boundaries in the input sequence, SEGBOT employs a pointer network and another recurrent neural network. This eliminated the need for hand-crafted elements in SEGBOT. More crucially, the problems of sparse boundary tags and variable size output vocabulary are naturally handled by SEGBOT. In two tasks, document-level topic segmentation and sentence-level EDU segmentation, SEGBOT surpasses the state-of-the-art models in our testing. Building on the results of SEGBOT, further suggest a hierarchical attention model for sentence-level sentiment analysis as a downstream application. Sentence-level sentiment analysis using the hierarchical model may fully utilize both word-level and EDU-level information at the same time. It may specifically efficiently utilize EDU-level data, such as the internal characteristics of EDUs, which are not fully stored in word-level features. According to experimental findings, our hierarchical model outperforms previous state-of-the-art performance on the Stanford Sentiment Treebank and Movie Review benchmarks.

Utilized a GOP Debate Twitter dataset, and conducted sentiment analysis by Gopalakrishnan and Salem [44] in 2020. Six distinct parameter-reduced slim variants of the LSTM model (slim LSTM) are offered to enhance training efficiency and decrease computational cost and time. Applying the dataset to both models, assess them. These two LSTM models' performances are compared to that of the conventional LSTM model. Additional research is done on the impact of bidirectional LSTM layers. In addition to determining which hyperparameter combination is optimal for each LSTM model, the effort includes a study to determine the optimal architecture. Tables 4 and 5 demonstrate the research gaps of aim, method, advantage, disadvantage, and Findings from the existing works of RNN.

Architectural overview

Unlike feedforward networks that treat each input independently, RNNs maintain a recurrent connection where the output from the previous time step is fed back into the model as part of the current input. This enables RNNs to model temporal dependencies across words and phrases in text. Formally, the hidden state at time step t is computed as:

$$h_t = f(W_h h_{t-1} + W_x x_t + b), \tag{3}$$

where x_t is the input vector (typically a word embedding), h_{t-1} is the previous hidden state, W_x and W_h are the weight matrices, b is the bias term, and $f(\cdot)$ is a nonlinear activation function such as tanh. The final sentiment representation h_T is passed to a fully connected or softmax layer for classification. Simplified architecture of a Recurrent Neural Network (RNN) for sentiment analysis is given in Fig. 6. The model processes input word embeddings

Table 4 Research gaps from the existing works of RNN

Author name	Aim	Methods	Advantage	Disadvantage	Findings
Tembhurne and Diwan [35]	Critically examine sequential deep neural networks (RNN and its architectural variants) in multimodal sentiment analysis	Literature review and critique	Sentiment analysis and emotion recognition for social media sites are in demand	Comprehensively covers the topic	Explores RNN and its architectural variants' sentiment analysis applications, problems, and methods in multiple modalities
Basiri et al. [36]	Detect sentiment polarity with an Attention-based Bidirectional CNN-RNN Deep Model (ABCDM)	Proposal and implementation of ABCDM model; Five reviews and three Twitter dataset sentiment polarity detection evaluation	Extracts past and future contexts using bidirectional LSTM and GRU layers	Complex architecture may need plenty of computing	ABCDM excels at long review and short tweet polarity categorization
Pathak et al. [37]	Present a taxonomy of attributes for deep learning-based sentiment analysis and explain its role	Framework creation based on key traits; Deep learning sentiment analysis discussion and analysis	Provides a framework for understanding deep learning-based sentiment analysis traits	Deep learning models demand plenty of processing power and time to train	Explains the consequences of deep learning for sentiment analysis

Table 4 continued

Author name	Aim	Methods	Advantage	Disadvantage	Findings
Baliyan et al. [38]	Use Neural Machine Translation (NMT) to propose a labeling system and RNN-LSTM for sentiment classification of multilingual text data	RNN-LSTM sentiment classification and NMT labeling system proposal	Sentiment classification using RNN-LSTM	Translation depends on the labeled English dataset, and may not apply to all languages	Shows that NMT and RNN-LSTM can analyze multilingual text sentiment
Alharbi et al. [39]	Compare RNN and word embedding algorithms for mobile phone review sentiment analysis	Implementation and evaluation of deep learning and word embedding algorithms for mobile phone review sentiment analysis	Compares sentiment analysis deep learning algorithms with feature extraction approaches	Lacks creativity in developing sentiment analysis methods and frameworks	Finds GLRNN algorithms with FastText feature extraction to be most accurate for imbalanced datasets and LRNN for balanced datasets
Alhumoud and Wazrah [46]	Systematic literature review to categorize, appraise, and identify state-of-the-art RNN Arabic sentiment analysis studies	A systematic review of Arabic RNN-based sentiment analysis literature	Explains Arabic RNN sentiment analysis	RNN-based sentiment analysis alone; may not cover other methods	Identifies RNN-based Arabic sentiment analysis study differences in regions addressed, functionality, shortcomings, and datasets

Table 5 Research gaps from the existing works of RNN continue...

Author name	Aim	Methods	Advantage	Disadvantage	Findings
Wankhade et al. [4]	Discuss sentiment analysis methods, evaluate and compare them, examine their pros and cons, and identify problems to shape future research	Comparing sentiment analysis approaches and their uses	Comprehensively explains sentiment analysis methodologies and applications	Due to broad coverage, technical details may be lacking	Explains sentiment analysis methods, their problems, and future directions
Sachin et al. [41]	Survey deep learning methods for sentiment classification and analysis, then create baseline models for LSTM, GRU, Bi-LSTM, and Bi-GRU on an Amazon review dataset	Survey of deep learning methods; Amazon review dataset baseline model implementation and evaluation	Uses deep learning to analyze customer review sentiment	Non-deep learning sentiment analysis approaches may be excluded	Uses baseline models to evaluate customer review sentiment analysis
Cen et al. [42]	Discuss three deep learning networks used in IMDB movie reviews sentiment analysis, their categorization impacts, and CNN, RNN, and LSTM models	Introduce deep learning networks for movie review sentiment analysis; Compare CNN, RNN, and LSTM models on the IMDB dataset	Explains sentiment analysis in movie reviews using deep learning networks	Concentrates on deep learning models, and may not cover other sentiment analysis approaches	Compared to RNN and LSTM models, the CNN network classifies movie review sentiment well

Table 5 continued

Author name	Aim	Methods	Advantage	Disadvantage	Findings
Patel et al. [43]	Use RNN for movie review sentiment analysis, monitor classifier performance, and achieve high accuracy	RNN algorithm for movie review sentiment analysis; classifier performance measurement	Deep learning for sentiment analysis captures complicated text patterns	Limited to RNN algorithm, may not consider other techniques	RNN algorithm achieves 94.61% movie review sentiment analysis accuracy
Li et al. [16]	Propose an end-to-end segmentation model SEGBOT for document-level topic and sentence-level EDU segmentation	Proposed SEGBOT model for segmentation; Hierarchical attention model for sentence-level sentiment analysis based on SEGBOT results	SEGBOT solves variable size output vocabulary and sparse boundary tag concerns without human feature engineering	Many computational resources and training time are needed for neural models	Performs better than state-of-the-art document-level topic and sentence-level EDU segmentation models; Hierarchical attention model achieves state-of-the-art sentence-level sentiment analysis
Gopalakrishnan and Salem [44]	Use LSTM models to analyze GOP Debate Twitter dataset sentiment, propose parameter-reduced slim versions of LSTM (slim LSTM), evaluate slim LSTM performance, and compare with standard LSTM model	LSTM sentiment analysis on GOP Debate Twitter dataset; Proposal and evaluation of slender LSTM models with decreased parameters; Bidirectional LSTM Layers and LSTM Model Architecture Optimization	Uses LSTM models to analyze social media sentiment	Limited to LSTM models, may not evaluate other deep learning architectures	Performs Twitter dataset evaluation of parameter-reduced thin LSTM models; Analysis of Bidirectional LSTM Layers and optimal architecture and hyperparameters for LSTM models

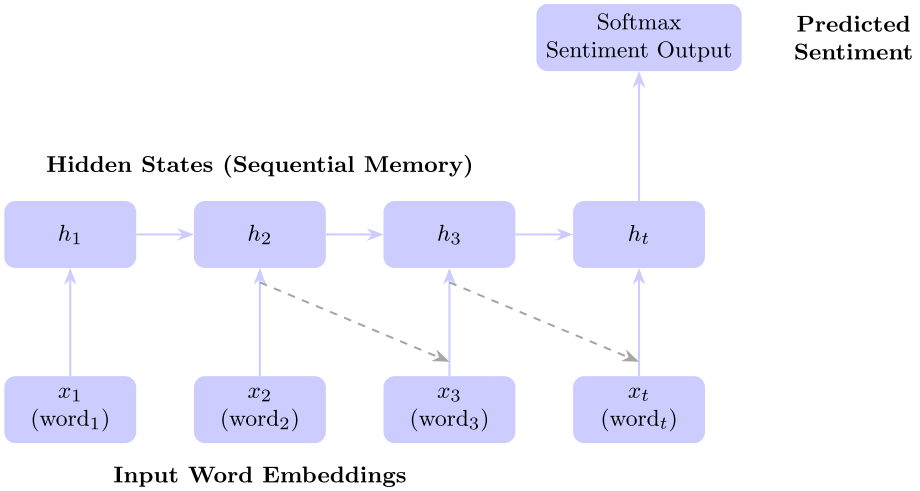


Fig. 6 Simplified architecture of RNN for sentiment analysis

sequentially, updating hidden states h_t at each time step to capture temporal dependencies, and produces a final sentiment prediction via a softmax output layer.

Analytical insights

RNNs capture sequential context effectively, making them suitable for sentiment classification tasks where word order and long-range dependencies influence meaning. For instance, in the sentence "The movie was boring at first but became amazing later," the sentiment reversal introduced by "but" requires sequential context modeling that RNNs handle naturally. Studies have shown that RNN-based models outperform bag-of-words and n-gram approaches by learning distributed representations that encode compositional semantics [16, 45].

However, standard RNNs face challenges such as the vanishing and exploding gradient problem, which limits their ability to retain information across long sequences. To address this, advanced variants like Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) were introduced. These models use gating mechanisms to control information flow, enhancing memory retention and improving sentiment prediction for long or complex sentences.

Limitations and future prospects

Despite their success, RNN-based architectures exhibit several limitations:

- *Sequential processing overhead* RNNs process tokens one by one, which restricts parallel computation and slows down training.
- *Context decay* Even with gating, maintaining dependencies across very long text remains challenging.
- *Limited bidirectionality* Standard RNNs only process context in one direction, which may overlook relationships dependent on subsequent words.

Recent hybrid models integrate bidirectional RNNs with attention mechanisms or pre-trained embeddings (e.g., BERT, RoBERTa) to improve contextual understanding while maintaining interpretability.

2.5 Sentiment analysis using deep belief network applications

The use of Deep Belief Networks (DBNs) in sentiment analysis tasks is examined in Sentiment Analysis Using Deep Belief Network Applications. Deep learning models called DBNs use several layers of hidden units to learn data representations that are hierarchical. DBNs are an excellent way to capture complex characteristics and patterns in text data for sentiment analysis, which allows for more sophisticated sentiment categorization. DBNs solve problems like data sparsity and feature extraction by utilizing unsupervised pre-training and fine-tuning techniques. Their use in sentiment analysis is widespread and demonstrates their effectiveness in reliably and accurately extracting sentiment from textual data. Examples of these applications include market sentiment prediction, customer feedback analysis, and social media monitoring.

Nair et al. [47] suggested how deep learning techniques outperform machine learning algorithms in terms of classification accuracy in 2021. The suggested techniques are assessed using specific datasets, such as reviews of Indian hotels, movies, and politics. Initially, deep learning methods were implemented, followed by the application of machine learning classifiers such as Naïve Bayes, Support Vector Machine, Random Forest, and logistic regression. The main contributions of several researchers are highlighted, with a primary focus on deep learning techniques. The algorithm's performance will be assessed based on factors such as F1 score, recall, accuracy, and precision. This work is novel because, instead of using RNN and LSTM together, we used a Deep Chrome Convolutional Neural Network for sentiment analysis. The results are impressive, outperforming other state-of-the-art algorithms in terms of accuracy, with 92.76% and 90.76% for movie and hotel reviews, respectively.

Agarwal and Gupta [48] proposed context-aware sentiment categorization using deep learning. Using the EfficientNetB-7 deep learning framework, we generated captions for the input image and then classified the emotions in the captions using machine learning. We first employ several synthetic and real-time image datasets to balance the data, and then we do preprocessing and normalization. Next, a successful module implementation for feature extraction and selection was completed using convolutional and pooling layers. Despite this, it generates the caption for the matching images. TF-IDF, lemmas, dependency, and correlational features are only a few of the several feature extraction and selection NLP techniques that are used to classify the sentiment label using the attention model and greedy strategy.

Deep learning is an effective technique that helps manage a lot of changing data was explored by Kale et al. [49] in 2023. The manufacturing sector offers a wealth of opportunities for data analysis and produces significant amounts of information every day. The majority of intelligent systems have been used to anticipate tool conditions; yet, descriptive analytics for on-board pattern recognition has to be investigated. We present the creation of a Deep Belief Network (DBN) to try and identify the variance in milling operation causing tool failures. The network uses real-time image-based vibration signals to distinguish six different tool conditions five of which are malfunctioning and one of which is healthy. Using datasets gathered while taking into account a variety of input parameters, the model was created, trained, tested, and validated.

In 2021, some of the earlier studies on sentiment analysis and the developments in text classification have been developed by Rajpoot et al. [50]. Due to the Internet's rapid growth and the volume of information available online, technology that can intelligently identify textual input is becoming increasingly important to the information retrieval process. The public can access the internet's response based on the query entered in the search box. Thus, several researchers are investigating the potential of text mining. Sentiment analysis is one of the most often used methods for analyzing user ideas and feelings. As online communication has become the fastest-growing means of thought expression, advances in text categorization have been made to enhance sentiment analysis.

In 2023, Ainapure et al. [51] examined Indian citizens' opinions regarding the COVID-19 epidemic and vaccine campaign through an analysis of text messages shared on Twitter. Deep learning and lexicon-based methods were used to categorize the sentiments. Using VADER and NRCLex, a lexicon-based method was employed to categorize the tweets' polarity. Using Bi-LSTM and GRU approaches, a recurrent neural network was trained on the COVID-19 dataset, attaining 92.70% and 91.24% accuracy. Using Bi-LSTM and GRU, the vaccine tweet classification yielded 92.48% and 93.03% accuracy rates, respectively. In the event of future pandemic outbreaks, the models that have been constructed can help politicians and healthcare professionals make the best choices.

Natural language processing (NLP) is the automated use of various approaches to represent human language as demonstrated by Etaiwi et al., [52] in 2021. Enhancements to natural language processing applications are crucial and can be achieved through a range of techniques, including word embedding, deep neural networks, and graphs. One of the main functions of sentiment analysis is sentiment classification, which aims to automatically categorize opinionated text as positive, negative, or neutral. This paper examined sentiment analysis techniques that have concentrated on deep learning over the last five years. Table 6 demonstrates the research gaps of aim, method, advantage, disadvantage, and Findings from the existing works of DBN.

Architectural overview

A DBN is composed of multiple stacked Restricted Boltzmann Machines (RBMs), where each RBM learns to represent the probability distribution of its inputs. The visible layer v represents observed input features (e.g., word embeddings), and the hidden layer h captures latent representations. The joint distribution of the visible and hidden units is defined as:

$$P(v,h) = \frac{1}{Z} e^{-E(v,h)}, \quad E(v,h) = -v^T W h - b^T v - c^T h, \tag{4}$$

where W is the weight matrix between the visible and hidden units, b and c are bias vectors, and Z is the partition function ensuring normalization. During pretraining, each RBM is trained independently using contrastive divergence to minimize reconstruction error. After unsupervised pretraining, the entire DBN is fine-tuned in a supervised manner using backpropagation for sentiment classification. Simplified architecture of a Deep Belief Network (DBN) for sentiment analysis is given in Fig. 7. The model consists of stacked Restricted Boltzmann Machines (RBMs) trained layer-wise in an unsupervised manner, followed by supervised fine-tuning for sentiment classification.

Table 6 Research gaps from the existing works of DBN

Author name	Aim	Methods	Advantage	Disadvantage	Findings
Nair et al. [47]	Assess deep learning's sentiment analysis performance against machine learning using diverse datasets	Compared machine learning classifiers (Naïve Bayes, SVM, Random Forest, logistic regression) and deep learning algorithms using movie, hotel, and political review datasets, evaluating accuracy, precision, recall, and F1 score metrics	Enhanced classification accuracy with deep learning	Beyond RNN and LSTM, proposed architectures lack innovation	Excellent accuracy (92.76% for movie reviews, 90.76% for hotel reviews) using Deep Chrome Convolutional Neural Network
Agarwal and Gupta [48]	Deep learning for context-aware sentiment categorization in image captions	Following image captioning with EfficientNetB-7, machine learning emotion classification utilizing convolutional and pooling layers and NLP methods including TF-IDF, lemmas, dependency, and correlational features	Effective feature extraction and deep learning sentiment categorization	Captions may not accurately reflect image dataset mood and prejudice	Excellent sentiment label categorization using the attention model and NLP
Kale et al. [49]	Create a Deep Belief Network (DBN) for descriptive analytics to identify manufacturing tool problems	Development, training, testing, and validation of a Deep Belief Network (DBN) to detect milling tool failures utilizing datasets with various input parameters	An effective way to handle vast volumes of dynamic data improves tool condition prediction	Little discussion of how the model handles complex tool faults	Successful real-time image-based vibration signal classification of six tool situations

Table 6 continued

Author name	Aim	Methods	Advantage	Disadvantage	Findings
Rajpoot et al. [50]	Read about text categorization and sentiment analysis advances due to online information proliferation	Comparison of sentiment analysis and text classification studies to examine how the field has evolved in response to internet information and the requirement to effectively categorize textual input	Previous sentiment analysis and text classification advances	Does not provide new sentiment analysis methods or systems	Explains sentiment analysis and text classification evolution
Ainapure et al. [51]	Twitter text texts can reveal Indians' views on the COVID-19 pandemic and vaccination drive	Lexicon-based (VADER and NRCLex) and deep learning (Bi-LSTM and GRU) sentiment analysis on Twitter data on the COVID-19 epidemic and vaccination drive, emphasizing accuracy metrics	Deep learning/lexicon-based sentiment classification	Lexicon-based methods may miss sentiment nuance	Classified COVID-19 and vaccine tweet emotions accurately
Etaiwi et al. [52]	Review deep learning sentiment analysis methods from the previous five years	Review and analysis of deep learning sentiment analysis approaches during the past five years to identify research advances and trends	An analysis of deep learning sentiment analysis	Sentiment analysis without deep learning is limited	Highlights deep learning-based sentiment analysis advances and trends

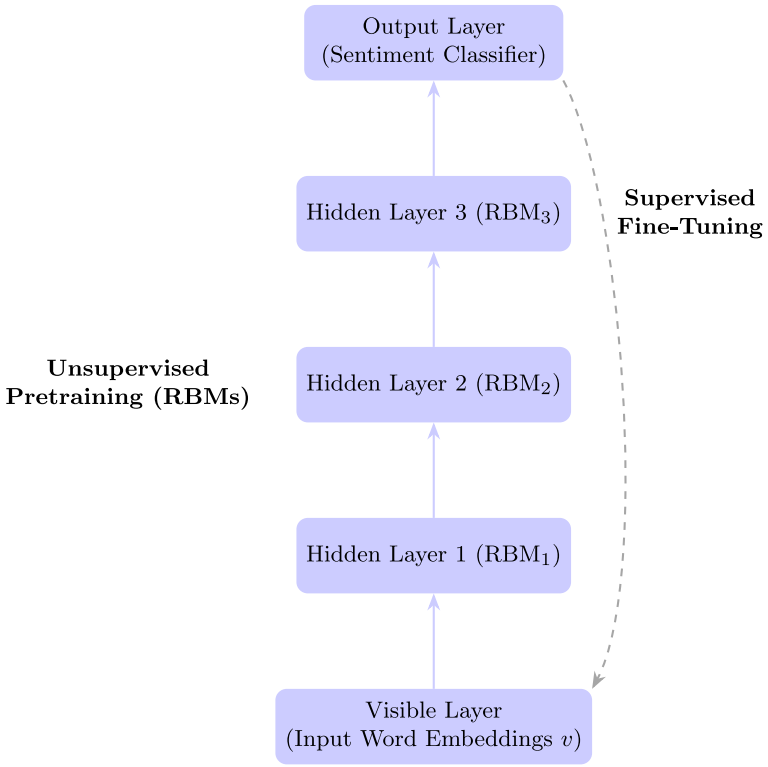


Fig. 7 Simplified architecture of DBN for sentiment analysis

Analytical insights

DBNs were among the first architectures to demonstrate that hierarchical unsupervised learning could improve sentiment analysis performance by capturing nonlinear relationships between linguistic features. They effectively learn distributed feature hierarchies – from local term frequency patterns in lower layers to global sentiment structures in higher layers. This deep feature representation helped early sentiment classifiers generalize better on limited labeled data [53].

However, unlike CNNs or RNNs, DBNs are generative in nature and lack sequence modeling capabilities. Consequently, they are less effective for tasks involving contextual dependencies, such as aspect-based or dialogue-level sentiment classification. Nevertheless, they remain historically significant as precursors to modern deep architectures and continue to inspire semi-supervised and transfer learning approaches in sentiment modeling.

Limitations and research trends

Despite their pioneering role, DBNs exhibit several limitations:

- *High computational cost* Layer-wise unsupervised pretraining and Gibbs sampling are time-intensive and scale poorly for large datasets.

- *Limited contextual awareness* DBNs treat text as feature vectors, ignoring sequential or syntactic structures.
- *Optimization challenges* Fine-tuning often leads to convergence instability due to vanishing gradients in deep stacks.

Recent research has explored integrating DBNs with word embeddings and autoencoders to enhance representation learning, as well as combining them with recurrent or transformer-based layers for hybrid sentiment modeling frameworks [54].

2.6 Transformer-based and domain-specific models

The transformer architecture has redefined the landscape of sentiment analysis by enabling deep contextual representation through self-attention mechanisms. While early implementations such as BERT achieved substantial success, recent transformer variants have demonstrated even greater robustness and semantic precision. State-of-the-art models including RoBERTa, DeBERTa, and XLNet introduce architectural refinements such as dynamic masking, disentangled attention, and permutation-based language modeling, resulting in more effective context modeling and generalization across domains. For instance, Cao [9] fine-tuned DeBERTa on IMDb and Amazon Polarity datasets, reporting over 93% accuracy and confirming its superiority over prior BERT and RoBERTa baselines. Similarly, Kaur et al. [10] compared multiple transformer architectures on financial discourse data (UK CBDC tweets) and found that RoBERTa achieved the highest F1-scores.

Furthermore, ensemble transformer models and hybrid approaches have recently emerged as promising directions. Vishal et al. [55] combined RoBERTa and DeBERTa within an ensemble framework for multi-label emotion detection, demonstrating that integrating transformer ensembles with lexical rules yields improved interpretability and accuracy. Domain-specific variants such as FinBERT and BioBERT extend this paradigm by tailoring pre-training corpora to specialized contexts, thereby improving performance in financial and biomedical sentiment tasks. Khurana [56] validated this trend by evaluating several transformer models on the SMILE Twitter dataset, showing that RoBERTa and DeBERTa outperform BERT and ALBERT for short-text sentiment classification. Finally, Gupta [57] emphasized the role of prompt engineering in enhancing domain adaptation for generic versus specialized NLP models, underscoring the growing integration between pre-training, fine-tuning, and task-specific prompting.

Overall, these studies confirm that transformer-based architectures—particularly RoBERTa, DeBERTa, and their domain-adapted counterparts—represent the current state of the art for sentiment analysis, providing superior contextual understanding, adaptability, and transferability compared to earlier CNN or RNN-based systems.

Comparative accuracy of deep learning models across benchmark sentiment analysis datasets (IMDb, SST-2, Twitter, and Amazon) is shown in Fig. 8. As discussed earlier in this section, transformer-based models such as RoBERTa and DeBERTa demonstrate superior performance over traditional architectures like CNN and RNN.

2.7 Problem statement

Although numerous algorithms have been developed for detecting emotions and opinions in text, the field of sentiment analysis continues to evolve rapidly, driven by the emergence of advanced deep learning architectures. Instead of proposing a new algorithmic framework,

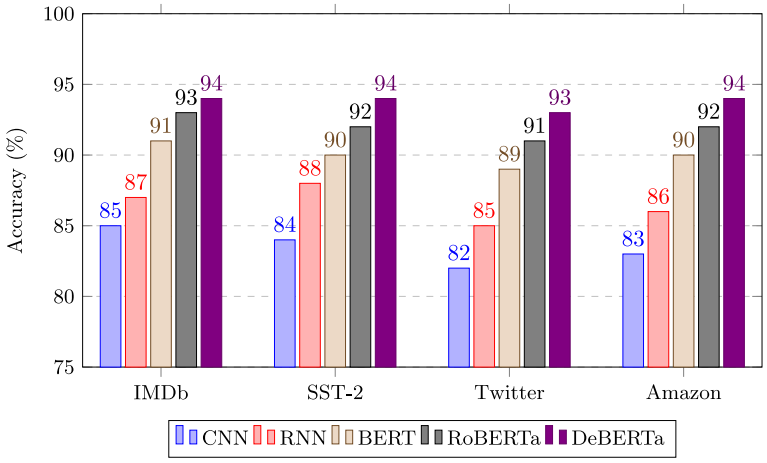


Fig. 8 Comparative accuracy of deep learning models across benchmark sentiment analysis datasets

the goal of this study is to provide a comprehensive and critical survey of existing deep learning approaches for sentiment analysis. The review focuses on synthesizing findings across different model categories–such as convolutional, recurrent, transformer-based, and hybrid architectures to evaluate their relative performance, generalization ability, and applicability across domains. By systematically examining methodological trends, benchmark datasets, and reported outcomes, this work seeks to establish an integrated understanding of current progress and to identify persisting research challenges and opportunities in sentiment analysis.

3 Mathematical modeling

This section presents the mathematical foundations of major deep learning architectures used in sentiment analysis. It formalizes the computational processes of CNNs, RNNs, and Transformer models, explaining how each architecture transforms textual data into structured sentiment representations through convolution, recurrence, and attention mechanisms.

3.1 Convolutional neural networks (CNN)

Convolutional neural networks are widely used in sentiment analysis for their ability to extract local n-gram features and hierarchical representations. The convolutional operation for a given feature map can be defined as:

$$h_{i,j}^{(k)} = f\left(\sum_{m,n} w_{m,n}^{(k)}\, x_{i+m,j+n} + b^{(k)}\right), \tag{5}$$

where $h_{i,j}^{(k)}$ represents the activation at position (i, j) for the kth feature map, $w_{m,n}^{(k)}$ are the convolutional kernel weights, $x_{i+m,j+n}$ is the input value at offset (m, n), and $b^{(k)}$ is the bias term. The function $f(\cdot)$ denotes a nonlinear activation (e.g., ReLU). Pooling and

flattening operations are subsequently applied to derive a fixed-length vector representation for classification.

3.2 Recurrent neural networks (RNN) and LSTMs

RNNs are particularly effective for modeling sequential dependencies in text. The hidden state at time step t is computed as:

$$h_t = f(W_h h_{t-1} + W_x x_t + b), \tag{6}$$

where x_t is the current input, h_{t-1} is the previous hidden state, and f denotes a nonlinear activation. However, standard RNNs struggle with long-term dependencies, which are mitigated by long short-term memory (LSTM) networks using gated mechanisms defined as:

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i), \tag{7}$$
$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f), \tag{8}$$
$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o), \tag{9}$$
$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_c x_t + U_c h_{t-1} + b_c), \tag{10}$$
$$h_t = o_t \odot \tanh(c_t), \tag{11}$$

where i_t, f_t, o_t represent the input, forget, and output gates respectively; $\odot$ denotes element-wise multiplication; and c_t is the cell state that preserves long-term information.

3.3 Transformer-based models

Transformer architectures such as BERT, RoBERTa, and DeBERTa rely on the self-attention mechanism to model global dependencies between words. The attention operation is expressed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \tag{12}$$

where Q, K, and V represent the query, key, and value matrices derived from the input embeddings, and d_k is the dimensionality of the key vectors. The multi-head attention mechanism extends this to multiple representation subspaces:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O, \tag{13}$$

with each attention head defined as:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V). \tag{14}$$

This formulation allows Transformer models to simultaneously capture syntactic, semantic, and contextual dependencies across varying positions in the text sequence.

4 Methodology

To ensure methodological rigor, the review was conducted using a systematic process encompassing database selection, article retrieval, relevance filtering, and analytical synthesis. This

approach enhances reproducibility, transparency, and academic integrity. The methodological novelty of this review lies in its systematic synthesis and analytical framework. Unlike descriptive surveys, this study follows a structured selection and evaluation protocol:

Literature was collected from Scopus, Web of Science, IEEE Xplore, and SpringerLink databases using predefined search terms.

Studies were filtered based on inclusion criteria such as publication date (2020–2025), peer-reviewed status, and focus on deep learning-based sentiment analysis.

Each model was evaluated using five analytical dimensions—architecture design, dataset characteristics, evaluation metrics, computational efficiency, and interpretability. This framework enables a balanced comparison of models and helps derive meaningful insights into their relative strengths, weaknesses, and application suitability.

4.1 Search strategy

A multi-database search was carried out using IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, Scopus, and Google Scholar. The time frame considered for this review was from January 2020 to October 2025 to capture recent developments in deep learning-based sentiment analysis. Keywords employed in the search included:

- "sentiment analysis" AND "deep learning"
- "text classification" AND ("CNN" OR "RNN" OR "Transformer" OR "LLM")
- "opinion mining" AND "natural language processing"
- "BERT" OR "LSTM" OR "GPT" AND "emotion detection"

Boolean operators and wildcard symbols were applied to expand the query scope. In addition, a snowballing technique was utilized to identify foundational papers through citations in highly referenced studies.

4.2 Selection criteria

The selection of studies adhered to predefined inclusion and exclusion criteria.

Inclusion criteria:

- Peer-reviewed articles published between 2020 and 2025.
- Research focusing on sentiment analysis utilizing deep learning architectures, including Large Language Models (LLMs).
- Articles presenting experimental validation or theoretical modeling.
- Full-text availability in English.

Exclusion criteria:

- Studies centered solely on traditional machine learning or lexicon-based methods.
- Articles lacking sufficient methodological detail or empirical evaluation.
- Duplicates retrieved from multiple databases.

4.3 Data extraction

Structured data extraction was conducted for each selected paper, capturing the following elements:

- Type and configuration of the deep learning model (e.g., CNN, BiLSTM, Transformer, LLM).

- Dataset characteristics and preprocessing strategies.
- Evaluation metrics employed (accuracy, F1-score, etc.).
- Nature of the contribution (architectural innovation, applied case study, comparative evaluation).
- Reported limitations and recommendations for future work.

Inconsistencies in interpretation were resolved through iterative document review and consensus.

4.4 Analytical framework

A structured comparative analysis was implemented by categorizing studies based on model type and evaluation methodology. Emphasis was placed on the following analytical dimensions:

- *Architectural depth* Complexity and novelty of neural network configurations, including transformer-based models and LLMs such as GPT, T5, and PaLM.
- *Linguistic nuance handling* Capacity to address sarcasm, irony, and multiple sentiment polarities.
- *Cross-domain robustness* Performance consistency across diverse datasets and contexts.
- *Computational efficiency* Training and inference time, scalability, and resource demands.
- *Interpretability* Transparency and comprehensibility of model decision-making processes.
- *Comparative performance* Effectiveness of LLMs relative to CNNs, RNNs, LSTMs, and conventional Transformer models, focusing on accuracy, generalization, and contextual understanding.

This synthesis served to identify prevailing trends, technological constraints, and underexplored avenues in the application of deep learning and large language models for sentiment analysis.

4.5 Scope of empirical review

It is important to note that this review does not include original experimental implementation or dataset benchmarking. Instead, it synthesizes empirical findings reported in prior studies that employed publicly available sentiment analysis datasets such as IMDB, SST-2, Twitter Sentiment140, Amazon Reviews, and Multi-Domain sentiment datasets. Evaluation metrics including accuracy, precision, recall, and F1-score were extracted from these studies to enable comparative assessment of deep learning architectures. The intention of this work is therefore integrative rather than experimental–providing a consolidated overview of empirical outcomes rather than proposing or executing new benchmarking procedures.

5 Performance analysis

Recent developments in deep learning, language processing, sentiment analysis, conversation systems, and language processing have greatly improved natural language comprehension in a variety of fields. RNNLG was first presented by Wen and Young [11], while DialTest was created for fault detection by Liu et al. [12]. The HAT model for response coherence was developed by Sun and Ding [13]. Discourse logic capture techniques were first presented by

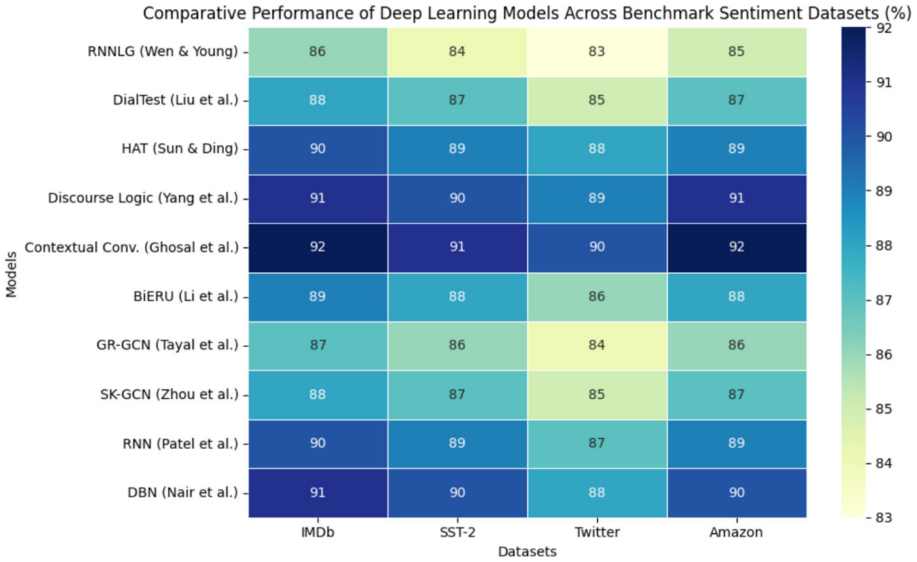


Fig. 9 Heatmap comparing the performance of various deep learning models across benchmark sentiment datasets

Yang et al. [14], while contextual understanding in conversation systems was investigated by Ghosal et al. [15]. For sentiment analysis, Li et al. [16] developed BiERU, while Tayal et al. [27] suggested GR-GCN for text processing. Zhou et al. [32] suggested SK-GCN for aspect-level sentiment categorization, whereas Patel et al. [43] carried out RNN-based sentiment analysis. The efficiency of DBNs was shown by Nair et al. [47], while sentiments on Twitter were examined by Ainapure et al. [51]. Summary of benchmark datasets commonly used for evaluating deep learning models in sentiment analysis are given in Table 7. Heatmap comparing the performance of various deep learning models is given in Fig. 9. This heatmap compares the performance of various deep learning models, including RNNLG, HAT, BiERU, GR-GCN, SK-GCN, and DBN, across benchmark sentiment datasets (IMDb, SST-2, Twitter, and Amazon). These developments highlight the adaptability and effectiveness of deep learning methods, including as RNNs, GCNNs, and DBNs, in comprehending and evaluating textual input in a variety of contexts and applications.

This review analyzes various deep learning models used for sentiment analysis and categorizes them based on their architecture and learning approach. We discuss supervised and unsupervised learning techniques, focusing on how models such as CNNs and RNNs process textual data. Additionally, we explore the role of transfer learning, particularly with pre-trained models like BERT and RoBERTa, in improving sentiment classification accuracy. We also assess benchmark datasets used in sentiment analysis research and the evaluation metrics employed to measure model performance.

6 Challenges

Despite their success, deep learning-based sentiment analysis models face challenges such as computational complexity, data scarcity, and domain adaptation issues. Handling sarcasm, irony, and ambiguous expressions remains a significant hurdle.

Table 7 Summary of benchmark datasets commonly used for evaluating deep learning models in sentiment analysis

Dataset	Domain / Source	Size	Classes / Labels	References
IMDb	Movie Reviews (Internet Movie Database)	50,000 samples	Positive / Negative	[58]
SST-2	Stanford Sentiment Treebank (Movie Snippets)	67,000 phrases	Binary Sentiment	[59]
Twitter	Social Media Posts (Tweets)	1.6 million posts	Positive / Negative / Neutral	[60]
Amazon	Product Reviews (Multi-domain)	100,000+ reviews	5-Class Rating Scale	[61]
Yelp	Business / Service Reviews	600,000 reviews	15 Star Ratings	[62]
SemEval	Twitter Sentiment Benchmark	30,000 posts	Positive / Negative / Neutral	[63]

- Improving the comprehensibility and elucidation of sentiment analysis models is essential for users to grasp the rationale behind model predictions and guarantee reliability.
- The task of addressing the need for sentiment analysis methods that can handle both textual data and visual and audio information is a notable obstacle, necessitating the creation of efficient multimodal methodologies.
- Maintaining the generalization of sentiment analysis models across many languages and domains is still a difficulty since models that are trained on certain datasets could not function well in other situations.
- To ensure justice, transparency, and accountability in the use of sentiment analysis systems, it is crucial to address ethical considerations and mitigate prejudices. This requires careful attention to issues such as algorithmic bias and fairness.
- The availability of high-quality labeled data in many areas and languages is crucial for the development of precise sentiment analysis models. However, the scarcity and quality concerns associated with data can present considerable obstacles, particularly in the case of languages with limited resources and specialized topics.
- Creating sentiment analysis models that can handle massive amounts of data in real-time or near-real-time, while still being scalable and efficient, continues to be a difficult task, especially as data volumes continue to increase rapidly.
- Maintaining performance levels when adapting sentiment analysis models to new domains or shifting situations poses challenges owing to variations in language usage, cultural subtleties, and domain-specific terminology.

Training and data considerations

Despite major advancements, several unresolved challenges persist in deep learning-based sentiment analysis, particularly concerning model training, data imbalance, and domain adaptation. Transformer models achieve exceptional accuracy only when effectively fine-tuned; however, optimal fine-tuning strategies and hyperparameter configurations remain inconsistent across studies. Cao [9] demonstrated that domain-aware fine-tuning of DeBERTa significantly enhances cross-domain generalization, while Kaur et al. [10] showed that RoBERTa's superior contextualization yields higher F1-scores when properly regularized for financial text. These results highlight the critical role of pre-training quality and task-specific adaptation in modern sentiment systems.

Data imbalance continues to impede model fairness and robustness, as minority sentiment classes (e.g., neutral or mixed emotions) are often underrepresented. Ensemble and augmentation techniques, such as those applied by Vishal et al. [55], help mitigate this issue but require further empirical standardization. Moreover, domain-specific sentiment lexicons–essential for nuanced understanding of specialized corpora like finance, healthcare, or social media–remain underutilized. Integrating such lexicons with contextual embeddings, as encouraged by Khurana [56], improves model interpretability and lexical grounding.

Finally, Gupta [57] emphasized that combining prompt engineering with domain-specific fine-tuning provides a scalable path to efficient retraining, enabling transformers to adapt to emerging linguistic patterns with minimal labeled data. Addressing these interconnected issues—balanced dataset design, informed fine-tuning, and lexicon integration—will be essential for building equitable, interpretable, and generalizable sentiment analysis systems.

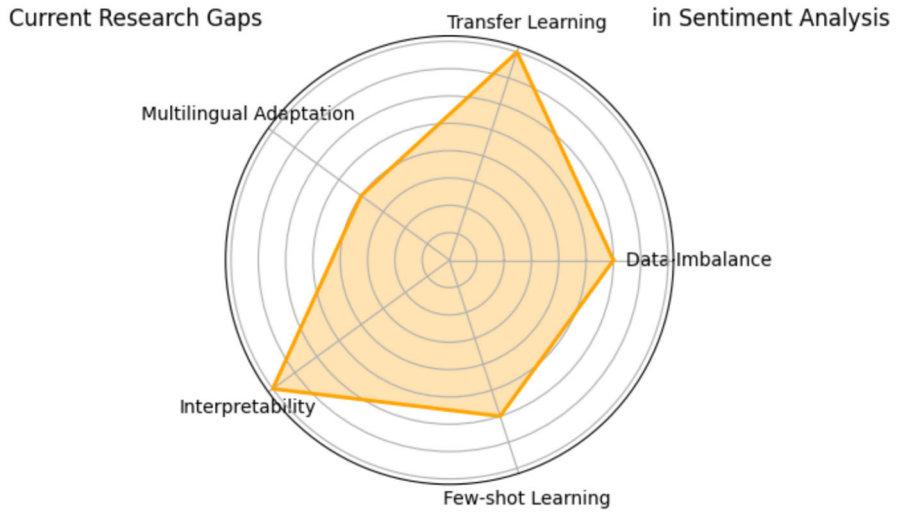


Fig. 10 Current Research Gaps in Sentiment Analysis

7 Research gaps

Despite notable advancements, several important research gaps remain in deep learning-based sentiment analysis. First, the field lacks standardized methodologies for fine-tuning transformer models across domains and tasks. Although recent studies such as Cao [9] and Kaur et al. [10] demonstrate performance gains through informed fine-tuning, optimal hyper-parameters, regularization strategies, and stability criteria remain underexplored, limiting reproducibility and practical deployment.

Second, research has not sufficiently addressed challenges related to data imbalance, label scarcity, and domain shift. While ensemble approaches proposed by Vishal et al. [55] and short-text evaluations by Khurana [56] show improvements through hybrid and domain-aware modeling, systematic benchmarks on imbalanced datasets and minority sentiment classes are still lacking. Moreover, domain-specific lexicons–essential for sectors such as finance, healthcare, and social media–are rarely integrated with contextual transformer embeddings, leaving significant room for improvement in interpretability and cross-domain transferability.

Third, emerging paradigms such as transfer learning, multilingual modeling, and zero-/few-shot sentiment adaptation remain underrepresented in current literature as shown in Fig. 10. Although multilingual contextual models (e.g., XLM-RoBERTa) and prompt-based adaptation techniques demonstrated by Gupta [57] offer robust cross-lingual and low-resource capabilities, comprehensive investigations into these models for sentiment analysis are scarce. The field lacks unified benchmarks to compare multilingual, cross-domain, and zero-shot sentiment performance, limiting progress toward globally scalable sentiment systems.

Finally, interpretability and ethical considerations, particularly model transparency, bias detection, and linguistic fairness, are insufficiently examined. Advanced transformer models provide state-of-the-art performance but often operate as "black-box" systems, complicating their use in high-stakes applications such as financial forecasting, healthcare triage, and policy

analysis. Addressing these research gaps will require integrated efforts in dataset design, algorithmic fairness, multilingual modeling, and interpretable deep learning frameworks.

8 Future directions

Fine-Tuned sentiment interpretation: Examine methods for capturing more complex expressions of sentiment than just positive, negative, or neutral labels. This could entail sentiment trajectory analysis across time, sentiment intensity analysis, or aspect-based sentiment analysis. Sentiment analysis in multiple modes: Develop techniques that combine textual, visual, and auditory modalities further to offer a more thorough comprehension of sentiment expressed in various media formats, including photographs, videos, and posts on social media. Analyzing sentiment cross lingual: To assist sentiment analysis activities in many linguistic contexts, look into sentiment analysis algorithms that can generalize across multiple languages, including low resource languages. Sentiment analysis using contextualization: Create models that are able to precisely represent sentiment in certain domains or scenarios, taking into account variables like user demographics, cultural variances, and situational contexts. Explicit and reliable models: Improve the interpretability, openness, and reliability of sentiment analysis models so that users can comprehend and have faith in the model's judgments while maintaining accountability and justice. Ethical issues and the mitigation of bias: Handle moral questions like prejudice, justice, and privacy in sentiment analysis systems, making sure that models are created and used ethically to reduce possible harm and increase social benefits. Robust and effective resolutions: Provide effective and scalable sentiment analysis methods that can handle massive amounts of data in real-time or almost real-time applications, allowing for deployment across a range of sectors and businesses. Person in the loop methods: Investigate techniques that enhance model performance, incorporate domain knowledge, and manage complex or unclear cases by fusing automated sentiment analysis with human experience. Domain adjustment and transfer education: Examine methods for exploiting transfer learning, domain adaptation, or semi-supervised learning methodologies to adapt sentiment analysis models to new domains or tasks with little labeled data. Continuous and lifelong learning: Create sentiment analysis models with the capacity to learn from and adjust to changing linguistic patterns, user preferences, and social shifts over time. This will allow the models to be improved and adjusted to constantly changing contexts.

Future research should focus on enhancing model interpretability, improving resource efficiency, and developing multilingual sentiment analysis frameworks. The integration of deep learning with other AI techniques, such as reinforcement learning and knowledge graphs, presents exciting opportunities for advancing sentiment analysis.

8.1 Transfer learning and multilingual adaptation

Transfer learning has become a pivotal paradigm in advancing sentiment analysis, allowing pre-trained language models to generalize effectively across tasks and domains with limited labeled data. Transformer-based models such as RoBERTa, DeBERTa, and XLNet enable efficient knowledge transfer through fine-tuning on task-specific datasets, substantially reducing the need for extensive retraining. Cao [9] demonstrated the effectiveness of fine-tuned DeBERTa for cross-domain sentiment classification, while Gupta [57] highlighted that prompt-based transfer learning can further enhance adaptability with minimal supervision.

Multilingual and cross-lingual sentiment analysis has gained increasing importance due to the global nature of online discourse. Models such as XLM-RoBERTa and mDeBERTa leverage multilingual pre-training to capture semantic relationships across languages, enabling robust sentiment prediction even for low-resource linguistic settings. Kaur et al. [10] underscored the advantages of multilingual contextualization for analyzing financial sentiment on diverse social media data. Additionally, zero-shot and few-shot adaptation methods–supported by large-scale instruction-tuned transformers–allow models to perform sentiment classification in unseen languages or domains without explicit retraining, a capability critical for scalable global applications.

Future research should focus on optimizing transfer learning protocols, enhancing cross-lingual representation quality, and developing benchmark datasets for evaluating zero-shot and few-shot sentiment analysis. Integrating multilingual pre-training with domain-specific fine-tuning and prompt engineering offers a promising path toward universal, adaptable, and equitable sentiment analysis frameworks.

9 Conclusion

Several prior reviews and empirical studies have highlighted the superiority of transformer-based and hybrid deep learning models in capturing contextual nuances. For instance, Basiri [36] demonstrated that an attention-based bidirectional CNN-RNN model outperformed traditional approaches on multiple sentiment benchmarks. Similarly, BERT significantly improves sentiment classification due to its deep bidirectional context modeling. The current study aligns with these observations and further integrates emerging perspectives on Large Language Models (LLMs), suggesting that they not only retain high performance but also generalize better across domains when fine-tuned. Thus, the conclusions presented here both reaffirm and extend recent advancements in sentiment analysis.

Deep learning has revolutionized sentiment analysis by providing more accurate and context-aware classification techniques. This paper reviewed key architectures, their advantages, and limitations while comparing them with traditional approaches. Although challenges remain, ongoing research continues to refine deep learning models for sentiment analysis. Future developments should aim to address existing limitations and enhance the adaptability of sentiment analysis techniques across different domains and languages.

Over the last ten years, sentiment analysis has become an essential component of natural language processing, enabling the extraction and classification of feelings from textual data from various sources. Sentiment analysis was widely used in areas such as politics, e-commerce, and online enterprises, highlighting its importance in understanding user ideas and feelings. Sentiment analysis approaches changed quickly as processing capacity increased, particularly with the introduction of deep learning algorithms. This article presented an extensive survey of the state-of-the-art in deep learning-based sentiment analysis, showcasing several architectures and methodologies used in the domain. The study clarified the possibility of using deep learning to improve sentiment analysis accuracy by analyzing the advantages and drawbacks of deep learning models as well as their effectiveness in sentiment classification tasks. As deep learning techniques were investigated further, sentiment analysis skills improved, and new research directions became available.

Author Contributions All authors contributed significantly to the development and completion of this review paper. Manoj Singh conceptualized the study, supervised the research, and led the writing of the manuscript. He also conducted the literature survey, analyzed deep learning architectures, and contributed to the interpretation

of results. Sanju Choudhary assisted in synthesizing methodologies, compiled performance evaluations, and contributed to reviewing emerging trends. Dr. Subhash Panwar reviewed, edited, and approved the final version of the manuscript.

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Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no Conflict of interest.

Human and/or animals participants This study does not involve any human participants or animals. No ethical approval was required.

Informed consent Not applicable.

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